

SIHAO LIU

Research Scientist, Architecture Research Group, NVIDIA Research

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RESEARCH INTERESTS

- Programmable spatial accelerators, hardware/software co-design, compilers, and on-chip interconnects.
- Agentic chip architecture and electronic design automation; independently reviewed and executable engineering workflows.

EDUCATION

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| 2026 | <ul style="list-style-type: none">• Ph.D. in Computer Science <p><i>Degree awarded September 2026. Advisor: Tony Nowatzki.</i>
Dissertation: Loom: Full-Stack Compilation and Automated Exploration of Heterogeneous Multi-Core Spatial Accelerator Systems.</p> | University of California, Los Angeles |
| 2021 | <ul style="list-style-type: none">• M.S. in Computer Science | University of California, Los Angeles |
| 2019 | <ul style="list-style-type: none">• B.Eng. in Electrical Engineering | Xi'an Jiaotong University |

PROFESSIONAL EXPERIENCE

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| 09/2026–
present | <ul style="list-style-type: none">• Research Scientist — NVIDIA Research
Architecture Research Group. Research on next-generation chip architectures and agentic systems for hardware design. |
| 06/2022–
10/2023 | <ul style="list-style-type: none">• Hardware Engineering Intern — SiFive
Memory management unit ownership; load/store and cache subsystem design and verification. Full-time summer and part-time academic-year appointments. |
| Summer
2021 | <ul style="list-style-type: none">• CPU Microarchitecture Intern — Futurewei Technologies
RISC-V vector CPU research and FPGA platform development. |

ACADEMIC EXPERIENCE

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| 2019–2026 | <ul style="list-style-type: none">• Doctoral Researcher — UCLA PolyArch Research Group |
| 2018 | <ul style="list-style-type: none">• Undergraduate Researcher (summer) — UCLA PolyArch Research Group |
| 2017 | <ul style="list-style-type: none">• Visiting Scholar (summer) — University of Lausanne
Neuroheuristic Research Group. Supervisor: Alessandro E. P. Villa. |

RESEARCH PROJECTS AND OPEN SOURCE

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| 2026 | <ul style="list-style-type: none">• Loom. A compiler and exploration framework for heterogeneous multi-core spatial accelerators, connecting a shared hardware model, compilation, simulation, and reproducible evaluation. |
| Open
source | <ul style="list-style-type: none">• Humanize / Humanfia. I created Humanize 1 at UCLA: an implementation and independent-review loop for coding agents. I contribute to Humanfia, where we extend this approach to longer engineering tasks. |
| 2025–2026 | <ul style="list-style-type: none">• OverNoC / BenchNoC. Characterization of hardened FPGA networks-on-chip and hierarchical network overlays for multi-chiplet reconfigurable systems. |
| 2020–2022 | <ul style="list-style-type: none">• DSAGEN / OverGen. Automated generation of programmable spatial architectures and FPGA overlays, bringing compilation, hardware implementation, and design-space exploration into one flow. |

CONFERENCE PUBLICATIONS

* Equal contribution. Preprints and the dissertation are listed separately.

- FCCM 2025 • H. Ke, **S. Liu**, L. Guo, Z. He, L. Song, S. Basalama, Y. Chi, T. Nowatzki, J. Cong. NoH: NoC Compilation in High-Level Synthesis.
- MICRO 2022 • **S. Liu**^{*}, J. Weng^{*}, D. Kupsh, A. Sohrabizadeh, Z. Wang, L. Guo, J. Liu, M. Zhulin, R. Mani, L. Zhang, J. Cong, T. Nowatzki. OverGen: Improving FPGA Usability through Domain-Specific Overlay Generation. **Best Paper Runner-up**.
- HPCA 2022 • Z. Wang, J. Weng, **S. Liu**, T. Nowatzki. Near-Stream Computing: General and Transparent Near-Cache Acceleration.
- ISCA 2021 • V. Dadu, **S. Liu**, T. Nowatzki. PolyGraph: Exposing the Value of Flexibility for Graph Processing Accelerators. **IEEE Micro Top Picks 2022**.
- ISCA 2020 • J. Weng^{*}, **S. Liu**^{*}, V. Dadu, Z. Wang, P. Shah, T. Nowatzki. DSAGEN: Synthesizing Programmable Spatial Accelerators. **IEEE Micro Top Picks 2021, Honorable Mention**.
- HPCA 2020 • J. Weng, **S. Liu**, Z. Wang, V. Dadu, T. Nowatzki. A Hybrid Systolic-Dataflow Architecture for Inductive Matrix Algorithms.
- MICRO 2019 • V. Dadu, J. Weng, **S. Liu**, T. Nowatzki. Towards General Purpose Acceleration by Exploiting Common Data-Dependence Forms. **IEEE Micro Top Picks 2020**.
- MICRO 2019 • A. Sharifian, R. Hojabr, N. Rahimi, **S. Liu**, A. Guha, T. Nowatzki, A. Shriraman. μ IR: An Intermediate Representation for Transforming and Optimizing the Microarchitecture of Application Accelerators.
- IUS 2017 • S. Zhang, S. Shang, Y. Han, R. Xu, **S. Liu**, L. Zhang, S. Wu, G. Niu, S. Wang, M. Wan. Monitoring of Microwave Ablation in Porcine Liver Using Ultrasonic Nakagami Imaging.

JOURNAL PUBLICATIONS

- Micro 2022 • J. Weng, **S. Liu**, D. Kupsh, T. Nowatzki. Unifying Spatial Accelerator Compilation With Idiomatic and Modular Transformations.
IEEE Micro, 42(5), 59-69.
- Micro 2022 • V. Dadu, **S. Liu**, T. Nowatzki. Systematically Understanding Graph Accelerator Dimensions and the Value of Hardware Flexibility.
IEEE Micro, 42(4), 87-96.
- Micro 2020 • V. Dadu, J. Weng, **S. Liu**, T. Nowatzki. Towards General-Purpose Acceleration: Finding Structure in Irregularity.
IEEE Micro, 40(3), 37-46.
- CAL 2019 • J. Weng, **S. Liu**, V. Dadu, T. Nowatzki. DAEGEN: A Modular Compiler for Exploring Decoupled Spatial Accelerators.
IEEE Computer Architecture Letters, 18(2), 161-165.
- TMI 2018 • S. Zhang, S. Shang, Y. Han, C. Gu, S. Wu, **S. Liu**, G. Niu, A. Bouakaz, M. Wan. Ex Vivo and In Vivo Monitoring and Characterization of Thermal Lesions by High-Intensity Focused Ultrasound and Microwave Ablation Using Ultrasonic Nakagami Imaging.
IEEE Transactions on Medical Imaging, 37(7), 1701-1710.
- IJH 2018 • S. Zhang, R. Xu, S. Shang, Y. Han, **S. Liu**, T. Xu, C. Gu, X. Zhu, G. Niu, M. Wan. In Vivo Monitoring of Microwave Ablation in a Porcine Model Using Ultrasonic Differential Attenuation Coefficient Intercept Imaging.
International Journal of Hyperthermia, 34(8), 1157-1170.

DISSERTATION, PREPRINTS, WORKSHOPS, AND BOOK CHAPTERS

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| Thesis
2026 | <ul style="list-style-type: none">• S. Liu. Loom: Full-Stack Compilation and Automated Exploration of Heterogeneous Multi-Core Spatial Accelerator Systems.
<i>Ph.D. dissertation, University of California, Los Angeles.</i> |
| arXiv
2025 | <ul style="list-style-type: none">• S. Liu, J. Ke, T. Nowatzki, J. Cong. Demystifying FPGA Hard NoC Performance.
<i>arXiv:2503.10861.</i> |
| WDDSA
2022 | <ul style="list-style-type: none">• J. Weng, S. Liu, D. Kupsh, T. Nowatzki. A Full-Stack Infrastructure for Automating Spatial Architecture Research.
<i>Workshop on Democratizing Domain-Specific Accelerators (WDDSA), co-located with MICRO.</i> |
| LATTE
2021 | <ul style="list-style-type: none">• J. Weng, V. Dadu, S. Liu, T. Nowatzki. Generality is the Key Dimension in Accelerator Design.
<i>Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE), co-located with ASPLOS.</i> |
| Chapter
2021 | <ul style="list-style-type: none">• Q. Zheng, S. Liu, A. E. P. Villa, A. Lintas. Unsupervised Analysis of EEG Signals Reveals Common Personality Traits During an Iterated Ultimatum Game.
<i>Advances in Cognitive Neurodynamics (VII), Springer, pp. 275-276.</i> |

HONORS AND AWARDS

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| 2022 | <ul style="list-style-type: none">• MICRO Best Paper Runner-up. OverGen. |
| 2022 | <ul style="list-style-type: none">• IEEE Micro Top Picks. PolyGraph. |
| 2021 | <ul style="list-style-type: none">• IEEE Micro Top Picks, Honorable Mention. DSAGEN. |
| 2020 | <ul style="list-style-type: none">• IEEE Micro Top Picks. SPU. |
| 2019 | <ul style="list-style-type: none">• Dean's Fellowship and University Fellowship. UCLA. |

ACADEMIC SERVICE

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| 2021 | <ul style="list-style-type: none">• Artifact Evaluation Committee, HPCA 2021. |
| 2021–2026 | <ul style="list-style-type: none">• Journal peer reviewer: ACM TACO, ACM TODAES, ACM TRETs, ACM TECS, IEEE TC, IEEE TCAD, IEEE TVLSI, IEEE CAL, IEEE JETCAS, Journal of Systems Architecture, Journal of Signal Processing Systems. |

TECHNICAL SKILLS

- **Programming:** C, C++, Rust, Python, Scala, MATLAB.
- **Hardware:** Chisel, Verilog, SystemVerilog; RTL design and verification; FPGA prototyping.
- **Compilers and simulation:** LLVM, MLIR, CIRCT, gem5, Chipyard, Rocket Chip, FireSim.
- **EDA tools:** Synopsys VCS, Design Compiler, IC Compiler; Verilator; AMD Vivado and Vitis.

RESEARCH IMPACT

913 citations; h-index 11; i10-index 11. Google Scholar snapshot, September 29, 2026.

EARLIER HONORS AND COMPETITIONS

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| 2017 | <ul style="list-style-type: none">• Finalist, Interdisciplinary Contest in Modeling. |
| 2017 | <ul style="list-style-type: none">• First Prize, China Big Data Modeling Contest. |
| 2016 | <ul style="list-style-type: none">• First Prize, Data Modeling Contest, XJTU. |
| 2016 | <ul style="list-style-type: none">• First Prize, Mathematical Modeling Contest, XJTU. |